Chapter 1

Bayesian Filter

§ 1.1 *Basic Settings*

Let

$$\boldsymbol{x}_t \in \mathbb{R}^{n_{\boldsymbol{x}}}$$

denote the *state* vector at time t, which represents the unknow quantity we want to infer.

At each time t, we have *observation*

$$\boldsymbol{z}_t \in \mathbb{R}^{n_{\boldsymbol{z}}},$$

which contains information about the hidden state $\boldsymbol{x}_t$, but may be noisy and uncertain. For convenience, deine the observation history up to time t as

$$\boldsymbol{z}_{1:t} = \{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_t\}$$

Similarly, suppose that a *control input* is available at each time t, denoted by

$$\boldsymbol{u}_t \in \mathbb{R}^{n_{\boldsymbol{u}}},$$

with the corresponding control history

$$\boldsymbol{u}_{1:t} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t\}.$$

Therefore, the available information evolves sequentially as

$$\boldsymbol{u}_1, \boldsymbol{z}_1, \boldsymbol{u}_2, \boldsymbol{z}_2, \boldsymbol{u}_3, \boldsymbol{z}_3, \ldots$$

whereas the states

$$\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots.$$

are not directly observed.

Since subsequent inference depends on the state, the objective is to characterize the uncertainty associated with the random variable $\boldsymbol{x}_t$ given all available information. This uncertainty is represented by the *belief state*

$$\mathrm{bel}(\boldsymbol{x}_t) = p(\boldsymbol{x}_t | \boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t}).$$

The belief state is therefore a probability distribution over the possible values of $\boldsymbol{x}_t$ conditioned on the available observation and control histories. It is also commonly referred to as the posterior state distribution, information state, or state of knowledge.

§ 1.2

Core Formula of Bayesian Filter

The key idea of the Bayesian filter is to represent uncertainty explicitly using the calculas of probability theory.

At each time step, the hidden state $\boldsymbol{x}_t$ evolves from the previous state $\boldsymbol{x}_{t-1}$ under the control input $\boldsymbol{u}_t$, while the observation $\boldsymbol{z}_t$ is generated from the corresponding state $\boldsymbol{x}_t$, as shown in Figure 1.1.

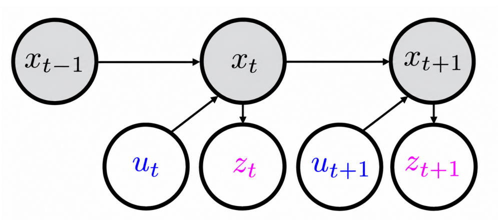

Figure 1.1 Graphical representation of the Bayesian filtering model

However, the belief, $p(\text{current sate}|\text{all past information})$, is intractable. This leads to the Markov assumption: future state conditionally independent of past control inputs and observations given present state. We therefore have

$$p(\boldsymbol{x}_t | \boldsymbol{x}_{t-1}, \boldsymbol{u}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) = p(\boldsymbol{x}_t | \boldsymbol{x}_{t-1}, \boldsymbol{u}_t),$$

$$p(\boldsymbol{z}_t | \boldsymbol{x}_t, \boldsymbol{u}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) = p(\boldsymbol{z}_t | \boldsymbol{x}_t).$$

Using Bayes' rule and the Markov assumptions,

$$
\begin{aligned}
\mathrm{bel}(\boldsymbol{x}_t) &= p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t}) = p(\boldsymbol{x}_t|\boldsymbol{z}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) && \leftarrow \text{Definition of belief} \\
&= \frac{p(\boldsymbol{x}_t, \boldsymbol{z}_t|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t})}{p(\boldsymbol{z}_t|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t})} && \leftarrow \text{Condition probability} \\
&= \eta_t p(\boldsymbol{x}_t, \boldsymbol{z}_t|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) && \leftarrow \text{Normalization constant} \\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) && \leftarrow \text{Bayes rule} \\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t) p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) && \leftarrow \text{Markov assumption} \\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t) \int p(\boldsymbol{x}_t, \boldsymbol{x}_{t-1}|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) \mathrm{d}x_{t-1} && \leftarrow \text{Marginalization} \\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t) \int p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) p(\boldsymbol{x}_{t-1}|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) \mathrm{d}x_{t-1} && \leftarrow \text{Probability chain rule} \\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t) \int p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) p(\boldsymbol{x}_{t-1}|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) \mathrm{d}x_{t-1} && \leftarrow \text{Unnecessary condition} \\
&= \underbrace{\eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t)}_{\text{correction}} \underbrace{\int p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{u}_t) \mathrm{bel}(\boldsymbol{x}_{t-1}) \mathrm{d}x_{t-1}}_{\text{prediction}}
\end{aligned}
$$

where η_t is a normalization constant,

$$
\eta_t^{-1} = p(\boldsymbol{z}_t|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = \int p(\boldsymbol{z}_t|\boldsymbol{x}_t) p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) \mathrm{d}\boldsymbol{x}_t
$$

and $\mathrm{bel}(x_0) = p(\boldsymbol{x}_0)$.

The Bayesian filter therefore provides a recursive update approach to calculate belief tractably consisting of two steps: prediction, which propagates the previous belief through the state transition model, and correction, which incorporates the latest observation through the observation model. The following gives the pseudocode of Bayesian filter.

Algorithm 1 Bayesian Filter

Require: Initial belief $\mathrm{bel}(\boldsymbol{x}_0) = p(\boldsymbol{x}_0)$, controls $\boldsymbol{u}_{1:T}$, observations $\boldsymbol{z}_{1:T}$, transition model $p(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{u}_t)$, and observation model $p(\boldsymbol{z}_t \mid \boldsymbol{x}_t)$

for $t = 1, 2, \ldots, T$ **do**
  $\overline{\mathrm{bel}}(\boldsymbol{x}_t) \leftarrow \int p(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{u}_t) \mathrm{bel}(\boldsymbol{x}_{t-1}) \mathrm{d}\boldsymbol{x}_{t-1}$ ▷ Prediction
  $\eta_t^{-1} \leftarrow \int p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) \overline{\mathrm{bel}}(\boldsymbol{x}_t) \mathrm{d}\boldsymbol{x}_t$ ▷ Normalization
  $\mathrm{bel}(\boldsymbol{x}_t) \leftarrow \eta_t p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) \overline{\mathrm{bel}}(\boldsymbol{x}_t)$ ▷ Correction
end for
return $\mathrm{bel}(\boldsymbol{x}_T)$

§ 1.3

Kalman Filter

The Kalman filter is a special case of the Bayesian filter in which the state transition and observation models are linear and all uncertainties are Gaussian.

1.3.1 Prediction Step

Suppose that the state transition is linear in the previous state $\boldsymbol{x}_{t-1}$ and control input $\boldsymbol{u}_t$:

$$\boldsymbol{x}_t = \mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t, \qquad \boldsymbol{q}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \qquad t \in \{1, \ldots, T\}.$$

Assume that the process noise $\boldsymbol{q}_t$ is independent of the previous state. Then

$$p(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{u}_t) = \mathcal{N}\left(\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t, \mathbf{Q}_t\right).$$

Suppose that the posterior state distribution at time $t-1$ is Gaussian:

$$p(\boldsymbol{x}_{t-1} \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) = \mathcal{N}\left(\boldsymbol{\mu}_{t-1|t-1}, \boldsymbol{\Sigma}_{t-1|t-1}\right),$$

where $\boldsymbol{\mu}_{t-1|t-1}$ and $\boldsymbol{\Sigma}_{t-1|t-1}$ denote the posterior mean and covariance based on all information available up to time $t-1$.

The prediction step of the Bayesian filter is

$$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = \int p(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{u}_t) p(\boldsymbol{x}_{t-1} \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) \mathrm{d}\boldsymbol{x}_{t-1}.$$

Since $\boldsymbol{x}_{t-1}$ and the independent process noise $\boldsymbol{q}_t$ are Gaussian, and $\boldsymbol{x}_t$ is an affine transformation of them, the predictive distribution is also Gaussian:

$$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = \mathcal{N}\left(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}\right).$$

Its mean is

$$\begin{aligned}
\boldsymbol{\mu}_{t|t-1} &= \mathbb{E}[\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}] && \leftarrow \text{Definition of predictive mean} \\
&= \mathbb{E}[\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t] && \leftarrow \text{State transition model} \\
&= \mathbf{F}_t \mathbb{E}[\boldsymbol{x}_{t-1}] + \mathbf{B}_t \boldsymbol{u}_t + \mathbb{E}[\boldsymbol{q}_t] && \leftarrow \text{Linearity of expectation} \\
&= \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \boldsymbol{u}_t && \leftarrow \text{Zero-mean process noise.}
\end{aligned}$$

Its covariance is

$$\begin{aligned}
\boldsymbol{\Sigma}_{t|t-1} &= \mathrm{Cov}(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) && \leftarrow \text{Definition of predictive covariance} \\
&= \mathrm{Cov}(\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t) && \leftarrow \text{State transition model} \\
&= \mathrm{Cov}(\mathbf{F}_t \boldsymbol{x}_{t-1} + \boldsymbol{q}_t) && \leftarrow \text{Deterministic control input} \\
&= \mathrm{Cov}(\mathbf{F}_t \boldsymbol{x}_{t-1}) + \mathrm{Cov}(\boldsymbol{q}_t) && \leftarrow \text{Independence of state and process noise} \\
&= \mathbf{F}_t \mathrm{Cov}(\boldsymbol{x}_{t-1}) \mathbf{F}_t^\top + \mathbf{Q}_t && \leftarrow \text{Covariance under linear transformation} \\
&= \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t .
\end{aligned}$$

Therefore, the prediction step of the Kalman filter is

$$\boldsymbol{\mu}_{t|t-1} = \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \boldsymbol{u}_t$$

and

$$\boldsymbol{\Sigma}_{t|t-1} = \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t .$$

1.3.2 Correction Step

Suppose that the observation model is also linear:

$$\boldsymbol{z}_t = \mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t, \qquad \boldsymbol{r}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \qquad t \in \{1, \ldots, T\},$$

where the observation noise $\boldsymbol{r}_t$ is independent of the state $\boldsymbol{x}_t$. Thus,

$$p(\boldsymbol{z}_t \mid \boldsymbol{x}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) = \mathcal{N}\left(\mathbf{H}_t \boldsymbol{x}_t, \mathbf{R}_t\right).$$

We first show that the predicted state $\boldsymbol{x}_t$ and the observation $\boldsymbol{z}_t$ are jointly Gaussian. From the observation model,

$$\begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{z}_t \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}_t \\ \mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{H}_t & \mathbf{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{r}_t \end{bmatrix}.$$

Before observing $\boldsymbol{z}_t$,

$$\boldsymbol{x}_t \sim \mathcal{N}\left(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}\right),$$

and

$$\boldsymbol{r}_t \sim \mathcal{N}\left(\mathbf{0}, \mathbf{R}_t\right).$$

Since $\boldsymbol{x}_t$ and $\boldsymbol{r}_t$ are independent Gaussian random vectors,

$$\begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{r}_t \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_{t|t-1} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{t|t-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_t \end{bmatrix}\right).$$

Because a linear transformation of a Gaussian random vector remains Gaussian, $\boldsymbol{x}_t$ and $\boldsymbol{z}_t$ are jointly Gaussian.

The predicted-state mean is

$$\boldsymbol{\mu}_{\boldsymbol{x}t} = \mathbb{E}[\boldsymbol{x}_t] = \boldsymbol{\mu}_{t|t-1}.$$

The predicted observation mean is

$$\boldsymbol{\mu}_{\boldsymbol{z}t} = \mathbb{E}[\boldsymbol{z}_t] = \mathbb{E}[\mathbf{H}_t\boldsymbol{x}_t + \boldsymbol{r}_t] = \mathbf{H}_t\mathbb{E}[\boldsymbol{x}_t] + \mathbb{E}[\boldsymbol{r}_t] = \mathbf{H}_t\boldsymbol{\mu}_{t|t-1}.$$

The predicted-state covariance is

$$\boldsymbol{\Sigma}_{\boldsymbol{x}\boldsymbol{x}t} = \mathrm{Cov}(\boldsymbol{x}_t) = \boldsymbol{\Sigma}_{t|t-1}.$$

The cross-covariance between the state and the observation is

$$\boldsymbol{\Sigma}_{\boldsymbol{x}\boldsymbol{z}t} = \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t) = \mathrm{Cov}(\boldsymbol{x}_t, \mathbf{H}_t\boldsymbol{x}_t + \boldsymbol{r}_t) = \mathrm{Cov}(\boldsymbol{x}_t, \mathbf{H}_t\boldsymbol{x}_t) + \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{r}_t) = \mathrm{Cov}(\boldsymbol{x}_t)\mathbf{H}_t^\top = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top.$$

Similarly,

$$\boldsymbol{\Sigma}_{\boldsymbol{z}\boldsymbol{x}t} = \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}.$$

Finally, the observation covariance is

$$\boldsymbol{\Sigma}_{\boldsymbol{z}\boldsymbol{z}t} = \mathrm{Cov}(\boldsymbol{z}_t) = \mathrm{Cov}(\mathbf{H}_t\boldsymbol{x}_t + \boldsymbol{r}_t) = \mathbf{H}_t\mathrm{Cov}(\boldsymbol{x}_t)\mathbf{H}_t^\top + \mathrm{Cov}(\boldsymbol{r}_t) = \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t.$$

Therefore,

$$\begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{z}_t \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_{t|t-1} \\ \mathbf{H}_t\boldsymbol{\mu}_{t|t-1} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{t|t-1} & \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \\ \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1} & \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t \end{bmatrix}\right).$$

The difference between the actual observation and its predicted mean is called the *innovation*:

$$\boldsymbol{\nu}_t = \boldsymbol{z}_t - \mathbb{E}[\boldsymbol{z}_t] = \boldsymbol{z}_t - \mathbf{H}_t\boldsymbol{\mu}_{t|t-1}.$$

Before observing $\boldsymbol{z}_t$, the innovation has zero mean:

$$\mathbb{E}[\boldsymbol{\nu}_t] = \mathbb{E}\left[\boldsymbol{z}_t - \mathbb{E}[\boldsymbol{z}_t]\right] = \mathbf{0}.$$

Its covariance is

$$\mathrm{Cov}(\boldsymbol{\nu}_t) = \mathrm{Cov}(\boldsymbol{z}_t) = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t.$$

We define the *innovation covariance*

$$\mathbf{S}_t = \underbrace{\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top}_{\text{uncertainty in the predicted observation}} + \underbrace{\mathbf{R}_t}_{\text{observation noise}}.$$

The two terms in $\mathbf{S}_t$ have distinct interpretations:

$$\underbrace{\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top}_{\text{uncertainty in the predicted observation}} + \underbrace{\mathbf{R}_t}_{\text{observation noise}}.$$

Since $\boldsymbol{x}_t$ and $\boldsymbol{z}_t$ are jointly Gaussian, conditioning on the newly observed $\boldsymbol{z}_t$ gives a Gaussian posterior:

$$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t}) = \mathcal{N}\left(\boldsymbol{\mu}_{t|t}, \boldsymbol{\Sigma}_{t|t}\right).$$

For jointly Gaussian random vectors, the conditional mean is

$$\begin{aligned}
\boldsymbol{\mu}_{t|t} &= \mathbb{E}[\boldsymbol{x}_t \mid \boldsymbol{z}_t] \\
&= \mathbb{E}[\boldsymbol{x}_t] + \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t)\mathrm{Cov}(\boldsymbol{z}_t)^{-1}\left(\boldsymbol{z}_t - \mathbb{E}[\boldsymbol{z}_t]\right) && \leftarrow \text{Conditional Gaussian formula} \\
&= \boldsymbol{\mu}_{t|t-1} + \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \mathbf{S}_t^{-1}\left(\boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}\right) && \leftarrow \text{Substitution of mean and covariance.}
\end{aligned}$$

Define the *Kalman gain*

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \mathbf{S}_t^{-1}.$$

The posterior mean can therefore be written compactly as

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t \boldsymbol{\nu}_t.$$

Thus, the new observation corrects the predicted mean by an amount proportional to the innovation $\boldsymbol{\nu}_t$, while the Kalman gain $\mathbf{K}_t$ determines how strongly the observation should influence the estimate.

For jointly Gaussian random vectors, the conditional covariance is

$$\begin{aligned}
\boldsymbol{\Sigma}_{t|t} &= \mathrm{Cov}(\boldsymbol{x}_t \mid \boldsymbol{z}_t) \\
&= \mathrm{Cov}(\boldsymbol{x}_t) - \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t)\mathrm{Cov}(\boldsymbol{z}_t)^{-1}\mathrm{Cov}(\boldsymbol{z}_t, \boldsymbol{x}_t) && \leftarrow \text{Conditional Gaussian formula} \\
&= \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \mathbf{S}_t^{-1}\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} && \leftarrow \text{Substitution of covariance terms} \\
&= \boldsymbol{\Sigma}_{t|t-1} - \mathbf{K}_t \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1}.
\end{aligned}$$

Equivalently,

$$\mathbf{\Sigma}_{t|t} = \left(\mathbf{I} - \mathbf{K}_t \mathbf{H}_t\right) \mathbf{\Sigma}_{t|t-1}.$$

Therefore, the correction step of the Kalman filter can be summarized as

$$\begin{aligned}
\boldsymbol{\nu}_t = \boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}, \quad \mathbf{S}_t = \mathbf{H}_t \mathbf{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t, \quad \mathbf{K}_t = \mathbf{\Sigma}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1}, \\
\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t \boldsymbol{\nu}_t, \quad \mathbf{\Sigma}_{t|t} = \left(\mathbf{I} - \mathbf{K}_t \mathbf{H}_t\right) \mathbf{\Sigma}_{t|t-1}.
\end{aligned}$$

Therefore, the new observation corrects the predicted state according to the discrepancy between the actual and predicted observations, while the magnitude of the correction depends on both the uncertainty of the state prediction and the observation noise. The complete Kalman filtering procedure is summarized in the following.

Algorithm 2 Kalman Filter

Require: Initial posterior mean $\boldsymbol{\mu}_{0|0}$, initial posterior covariance $\mathbf{\Sigma}_{0|0}$, controls $\boldsymbol{u}_{1:T}$, observations $\boldsymbol{z}_{1:T}$, state transition matrices $\mathbf{F}_{1:T}$ and $\mathbf{B}_{1:T}$, observation matrices $\mathbf{H}_{1:T}$, process noise covariances $\mathbf{Q}_{1:T}$, and observation noise covariances $\mathbf{R}_{1:T}$

```
for t = 1, 2, ..., T do
    μ_{t|t-1} ← F_t μ_{t-1|t-1} + B_t u_t            ▷ Predicted mean
    Σ_{t|t-1} ← F_t Σ_{t-1|t-1} F_t^⊤ + Q_t          ▷ Predicted covariance
    ν_t ← z_t − H_t μ_{t|t-1}                         ▷ Innovation
    S_t ← H_t Σ_{t|t-1} H_t^⊤ + R_t                   ▷ Innovation covariance
    K_t ← Σ_{t|t-1} H_t^⊤ S_t^{-1}                    ▷ Kalman gain
    μ_{t|t} ← μ_{t|t-1} + K_t ν_t                     ▷ Posterior mean
    Σ_{t|t} ← (I − K_t H_t) Σ_{t|t-1}                 ▷ Posterior covariance
end for
return {μ_{t|t}, Σ_{t|t}}_{t=1}^T of bel(x_t)
```